# Cross-Lot Generalization of Vision Foundation Models for Semiconductor Wafer Defect Classification

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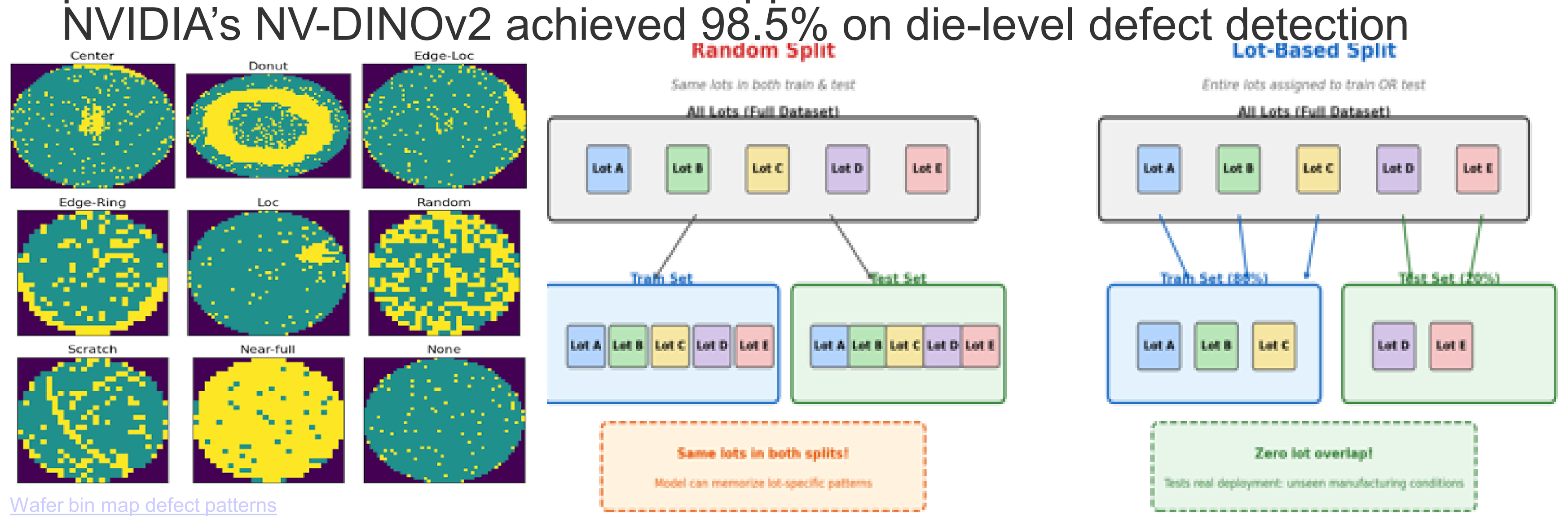
## Background

Semiconductor manufacturing requires accurate classification of spatial defect patterns on wafer bin maps (WBMs) (WM-811K dataset is the standard)

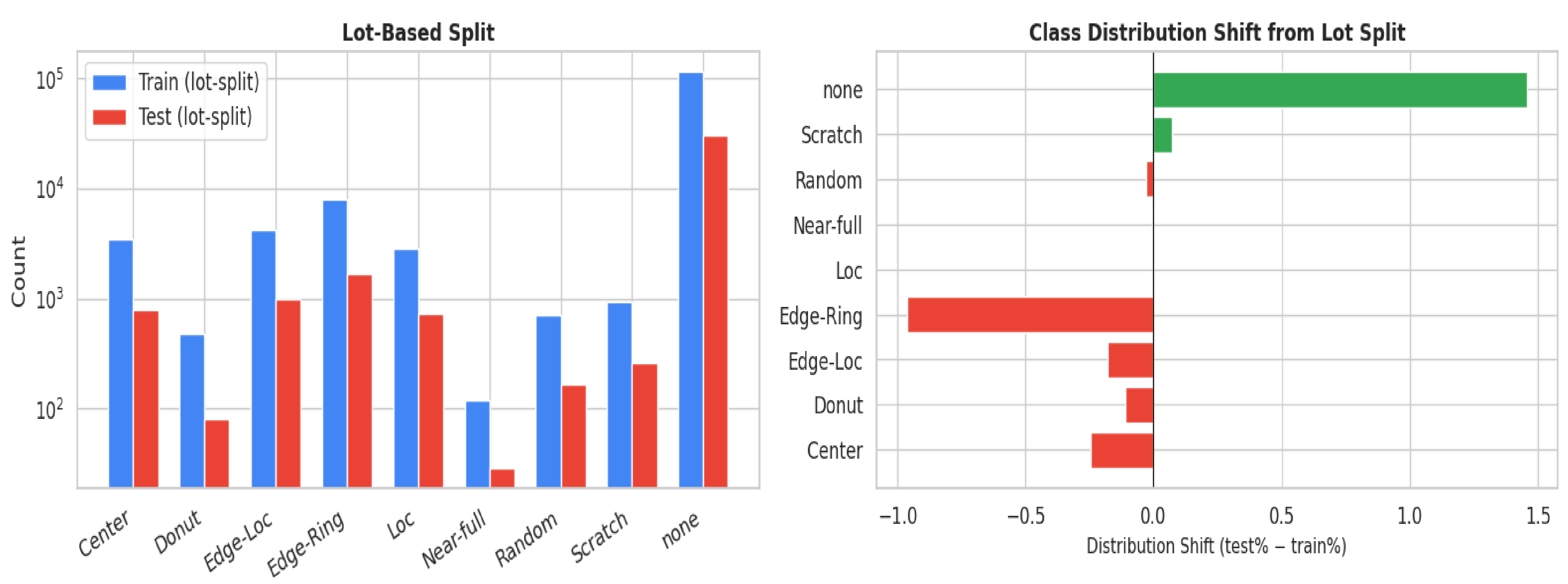
Traditional supervised CNNs achieve 97-99% accuracy on random train/test splits. This evaluation hides a critical deployment challenge:

- Models must generalize across manufacturing lots with different process conditions.

Vision foundation models pretrained on massive datasets offer a potential solution. Recent work supports this direction with NVIDIA's NV-DINOv2 achieved 98.5% on die-level defect detection




Lot-based splitting creates natural distribution shift — unlike random splits




## Objectives

(1)Compare frozen feature quality of various models for wafer defect classification using linear probing.

- Self-supervised (DINOv2),
- Language-supervised (CLIP)
- Supervised (ResNet-50)

(2)Determine how many labeled examples each model needs to reach production-quality accuracy.

(3)Test cross-lot generalization by splitting data by lot number rather than randomly, measuring whether foundation models transfer better across unseen manufacturing conditions.

## Approach/Research Methods

### Dataset and Preprocessing

WM-811K wafer bin maps converted to 224x224 RGB images.
(1) random 80/20 stratified split
(2) lot-based split where entire lots are assigned to train or test with zero overlap.

### Models Under Evaluation

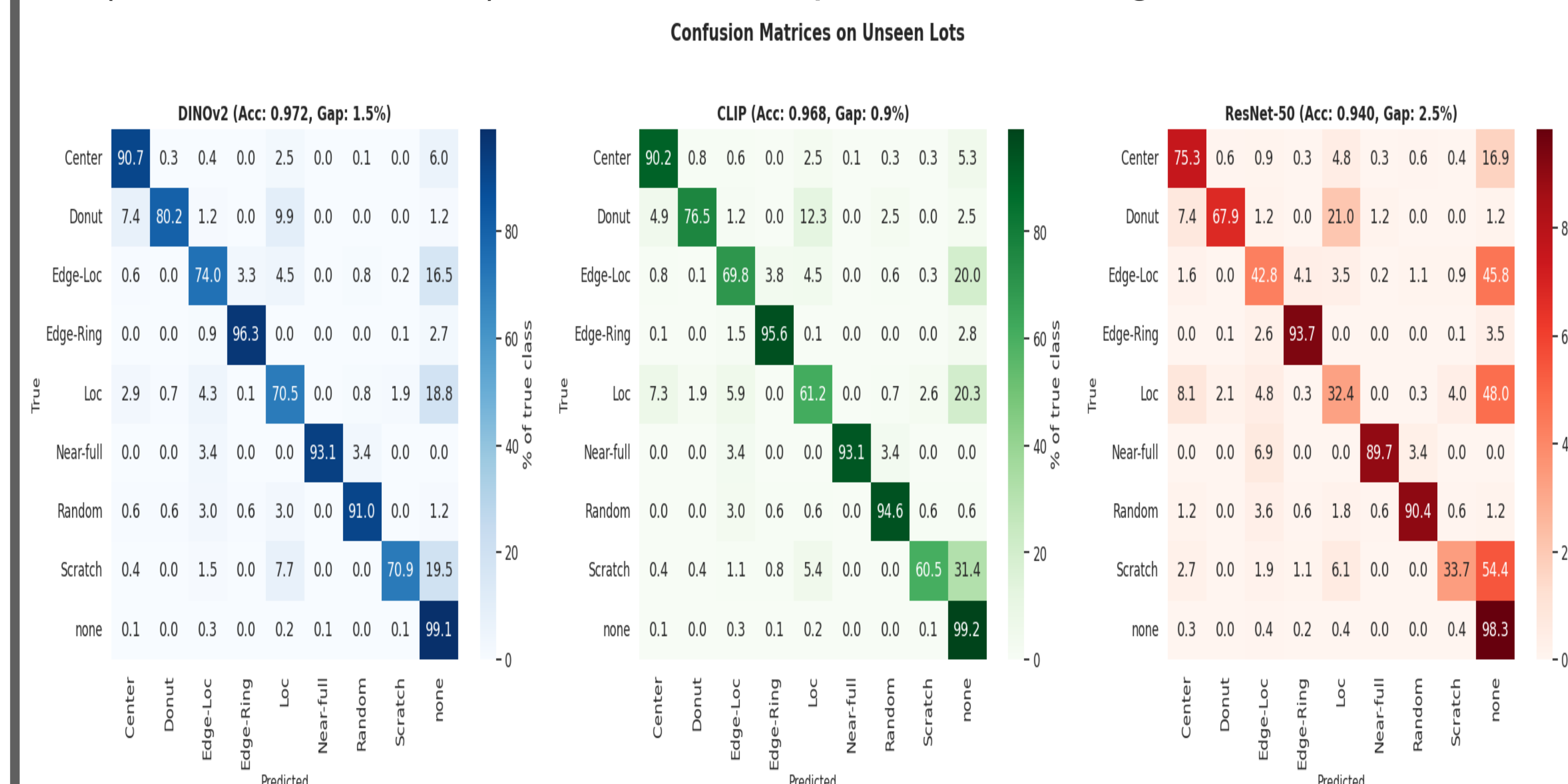
- DINOv2 ViT-L/14: Self-supervised, 304M params.
- CLIP ViT-L/14: Language-supervised, 428M params.
- ResNet-50: Supervised ImageNet baseline, 25.6M params.

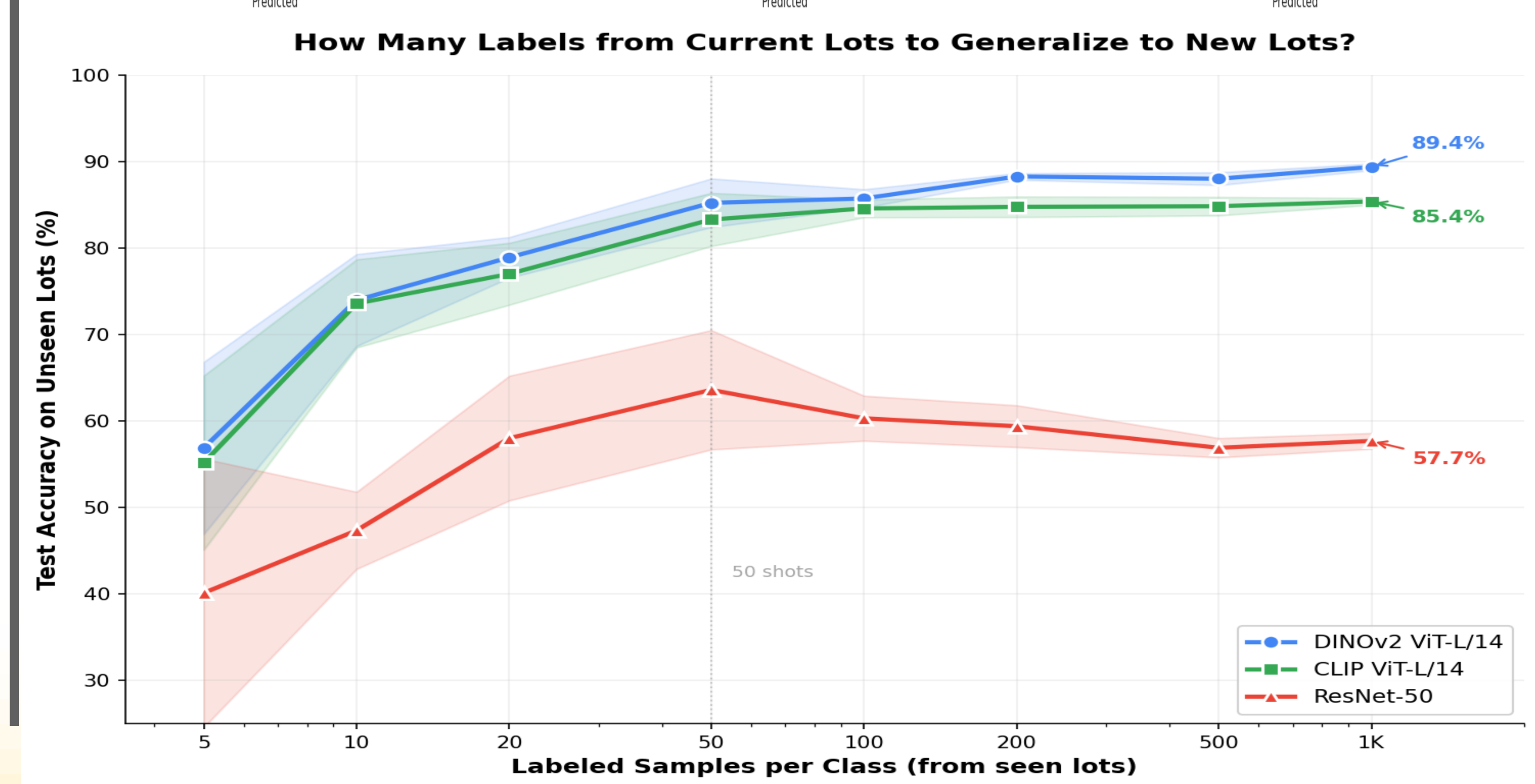
### Evaluation Protocol

- All models used as frozen feature extractors
- Linear probing
- Few-shot learning curves (5-1000 shots/class),
- Embedding quality (UMAP, silhouette scores, k-NN),

### Cross-Lot Generalization

Key experiment: split by lotName column to test whether models generalize across manufacturing conditions. The generalization gap (train acc - test acc) measures lot-specific overfitting.



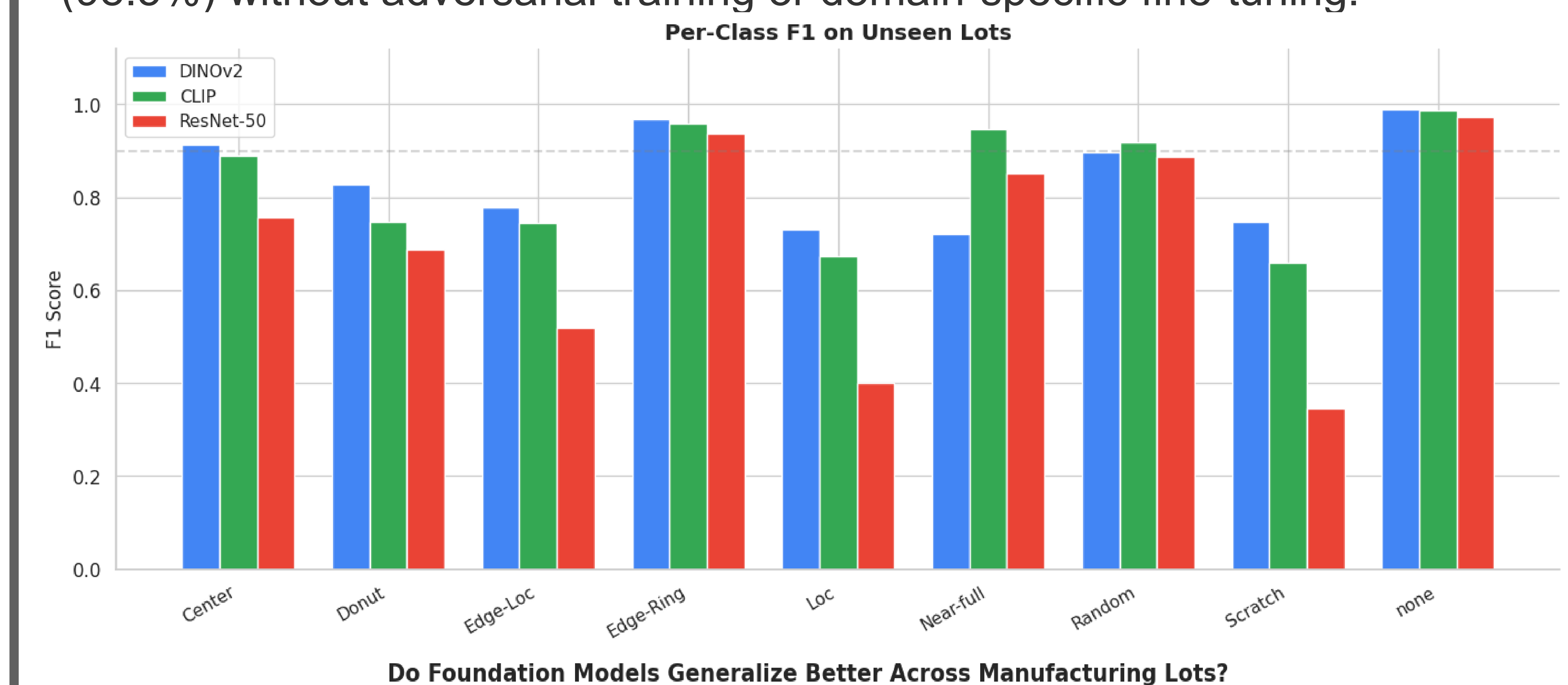

## Conclusions

**DINOv2 achieves highest linear probe accuracy (~97.2%)**
Self-supervised pretraining produces the most transferable frozen features, outperforming CLIP (~95.8%) and ResNet-50 (~94.1%).

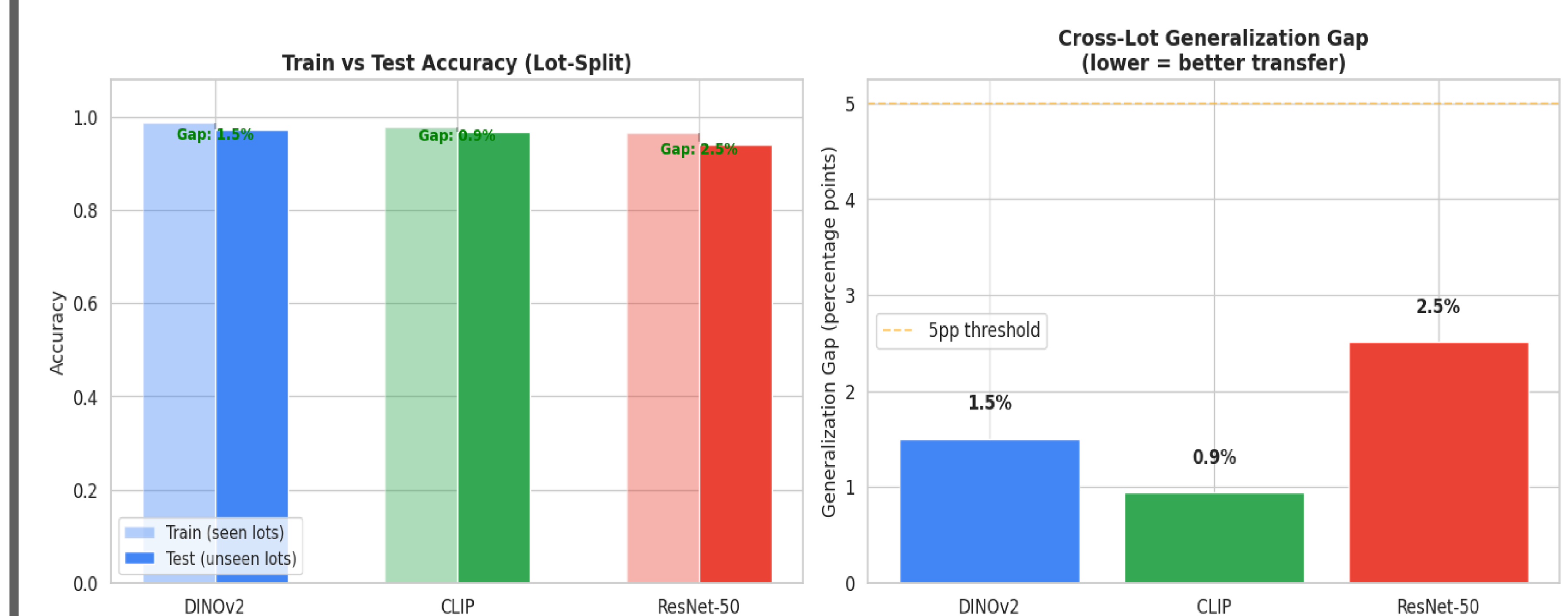
**Foundation models show smaller cross-lot gaps**
DINOv2 (3.4pp) and CLIP (4.3pp) degrade far less than ResNet-50 (8.8pp) on unseen lots.

**CLIP enables zero-shot defect classification**
CLIP classifies defects using only text descriptions, eliminating the labeling bottleneck for new fab lines.

**Off-the-shelf models rival custom domain adaptation**
Frozen DINOv2 approaches DTWAN (98.8%) and NV-DINOv2 (98.5%) without adversarial training or domain-specific fine-tuning.




Do Foundation Models Generalize Better Across Manufacturing Lots?




## References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